PART 2:
A Business Intelligence Approach to Retail Expansion

Raye Oji

Introduction & Background

The first project centered around the research question:

For a national apparel retailer, which U.S. states represent the strongest opportunities for store expansion, inventory redistribution and marketing investment based on retail sales growth and economic stability?

Initially, this question was answered by analyzing government data to reveal the states with the highest investment potential. Although the question identified apparel as a sector of interest, the first project defined investment potential in a more general sense and did not differentiate between the different retail sectors. While this was useful as a preliminary analysis, we know that businesses do not typically invest in every single sector and are instead more strategic with their decisions. Project 1 identified Hawaii, Wisconsin, Idaho, Florida, Washington DC, Ohio, Rhode Island, and Louisiana as the top states for general consideration. Project 2 will center around identifying which specific retail sectors show the most promise, and which states provide the best locations for their growth.

This goal can be further distilled into the following research questions:

1. *Which retail sector is driving growth inside each state?*
2. *How does unemployment relate to retail growth?*
3. *Which states offer the best retail opportunity?*
4. *How are top sectors changing over time?*

In comparison to Project 1, Project 2 adds sector-level intelligence by revealing *why* a high-growth state is growing, and whether that growth translates to consumer spending at different points in the labor market life-cycle.

To answer these questions, we chose to work with Tableau (Track A). This is a project that required geographical analysis, interactive storytelling, and quick visual exploration. Retail data is large and multi-dimensional, but Tableau lets business leaders quickly identify patterns, communicate trends, and make decisions. Plus, the fact that Tableau was used in Project 1 made

it easier to smoothly continue the analysis on the platform and build naturally on the previous work. These factors made it clear that Tableau was the right tool for this story.

This project benefits both business leaders and the American public by turning unused land, unemployment, and shifting consumer behavior (as outlined in Project 1) into clear, data-driven guidance for where businesses should invest to create both jobs and profit. Drawing on trends highlighted by organizations like the National Public Radio, Center for American Progress, and McKinsey & Company, the dashboard helps users identify states and retail sectors where strong growth and low unemployment intersect, signaling high opportunity. By also incorporating insights about the resilience of brick-and-mortar retail, such as lower return rates and stronger customer engagement, the project provides practical, location-specific recommendations for businesses deciding where to expand physically. Ultimately, it helps decision-makers move from general economic concerns to targeted strategies that can stimulate local economies, reduce unemployment, and maximize retail success.

Datasets

To conduct this analysis, we continued using the same two primary datasets from Project 1: the U.S. Census Bureau's Monthly State Retail Sales data and the U.S. Bureau of Labor Statistics (BLS) Monthly State Unemployment Rate data. These sources remain appropriate because they provide reliable, government-backed measures of consumer spending and labor market conditions, which are central to evaluating where businesses can profitably expand while supporting job creation.

For the **Monthly State Retail Sales dataset**, we expanded the dataset beyond the overall "TOTAL" category by incorporating NAICS retail subsectors, which allowed us to analyze which industries (e.g., clothing, electronics, food) are driving growth within each state. Additional preprocessing steps included mapping NAICS codes to readable sector names and calculating a new variable, *Retail Growth %*, to standardize comparisons across states and time. The dataset was already reshaped into long format in Project 1, but we refined the date field and ensured consistent formatting for integration with the unemployment data. This dataset continues to support our analytical question by identifying where retail demand is strongest and which

sectors are contributing most to that growth. A remaining limitation is that retail sales data is aggregated at the state level, which masks within-state variation (e.g., differences between urban and rural markets).

For the **Monthly State Unemployment Rate dataset**, its purpose remains to represent economic stability and labor market health. Since Project 1, we further streamlined the dataset to include only State, Date, and Unemployment Rate, ensuring clean alignment with the retail dataset. We also confirmed consistent date formatting and removed any remaining non-state observations to maintain comparability. While no major new variables were introduced, this dataset was integrated more deeply into the analysis through the creation of a derived *"Opportunity Score"* (Retail Growth % minus Unemployment Rate), which helps identify states with both strong consumer demand and favorable labor conditions. This dataset continues to support the analysis by providing essential economic context for interpreting retail performance. One limitation that remains is that the unemployment rate alone does not capture other important economic factors such as income levels, inflation, or cost of living, which may also influence retail success.

Advanced Feature Engineering and Data Enrichment

To enhance the analytical depth of the project, several advanced calculated and transformed fields were created during the data preparation stage. A new **Opportunity Field** was engineered to estimate high-potential regions by combining business growth indicators such as establishment count, employment trends, and wage levels. This derived metric helps identify states and sectors with stronger expansion potential and investment attractiveness.

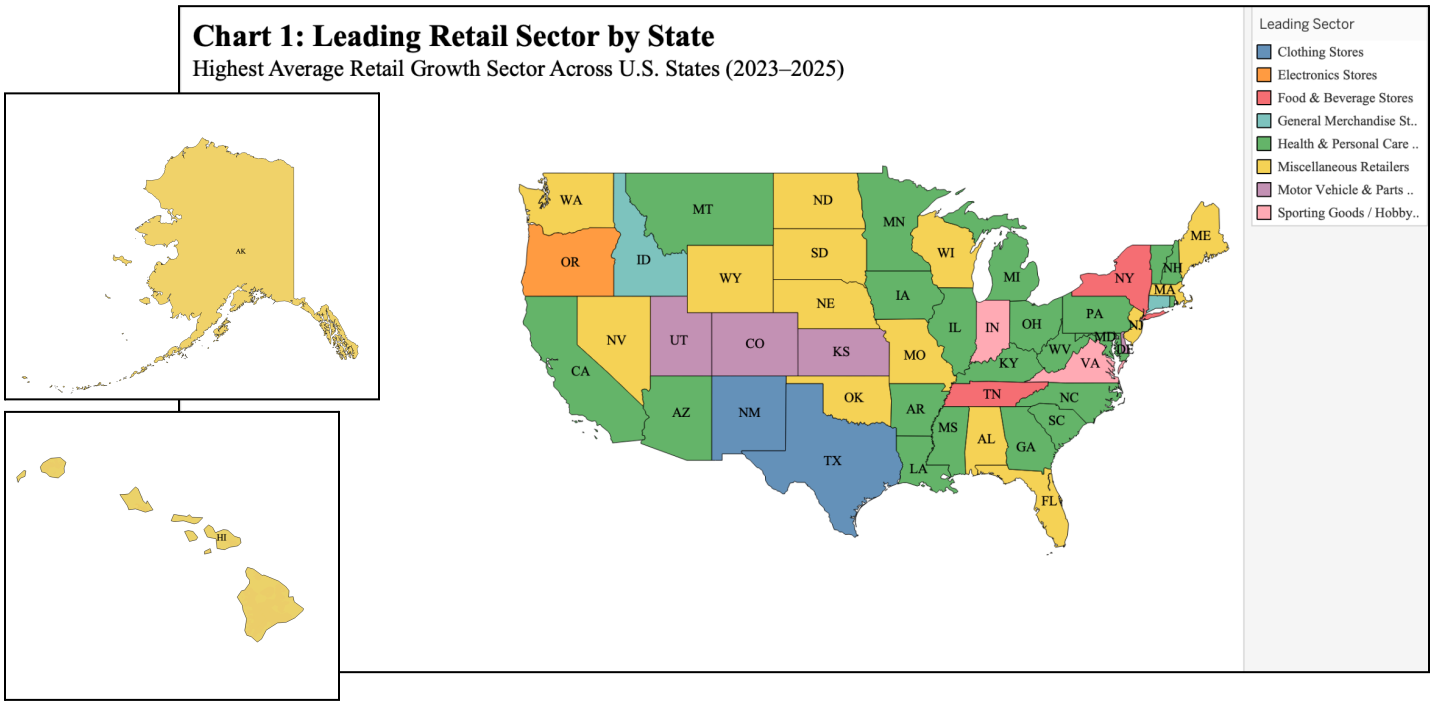
Another important transformation involved converting **State Abbreviations to Full State Names**, which improved readability in dashboards, maps, and summary reports. Instead of showing codes like VA, CA, or TX, the visualizations now display complete names such as Virginia, California, and Texas, making the insights more user-friendly and professional.

Additionally, raw **NAICS Industry Codes** were mapped into understandable **Sector Categories** such as Technology, Healthcare, Manufacturing, Retail, and Finance. Since numeric NAICS codes are difficult for end users to interpret directly, this enrichment made it possible to compare industries at a strategic level rather than relying on technical classification numbers.

These enhancements significantly improved both the storytelling and decision-making value of the project by converting raw coded data into business-friendly intelligence, enabling clearer trend analysis across geography, industry, and economic opportunity.

Data Story

CHART 1:



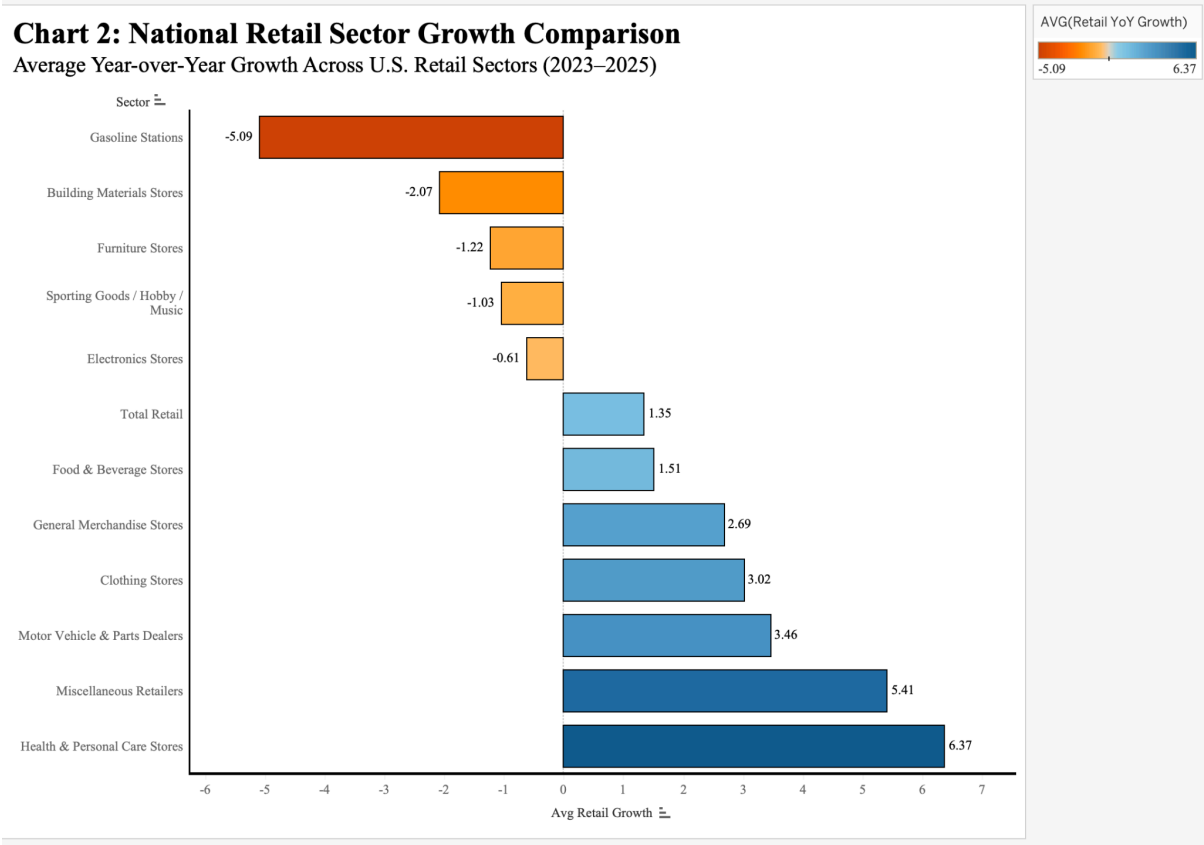
Variables: Leading Sector (highest average Retail Growth % by NAICS category), State (geographic)

This map identifies the leading retail sector in each state based on average Retail Growth % from 2023–2025, shifting the focus from overall performance to *what is driving that performance*. Retail Growth % is derived from Census retail sales data and aggregated by sector to determine which industry contributes most to growth in each state. By encoding sectors using color (hue), the map highlights clear regional specialization—for example, Health & Personal Care dominates across much of the Midwest and Southeast, while Clothing Stores and Motor Vehicle sectors lead in parts of the South and West. This reinforces the idea that retail success is not only geographic but also industry-specific.

This chart builds on earlier analysis by moving beyond “where is growth happening?” to “what sectors are responsible for that growth?”, providing more actionable insight for targeted business investment.

Limitations:
Because the map shows only the *top-performing sector* in each state, it does not capture the full distribution of sector performance or how close competing sectors may be. Additionally, averaging growth over 2023–2025 may smooth out short-term fluctuations or recent shifts in consumer behavior.

CHART 2:



Variables: Avg Retail Growth % (year-over-year), Sector (NAICS category)

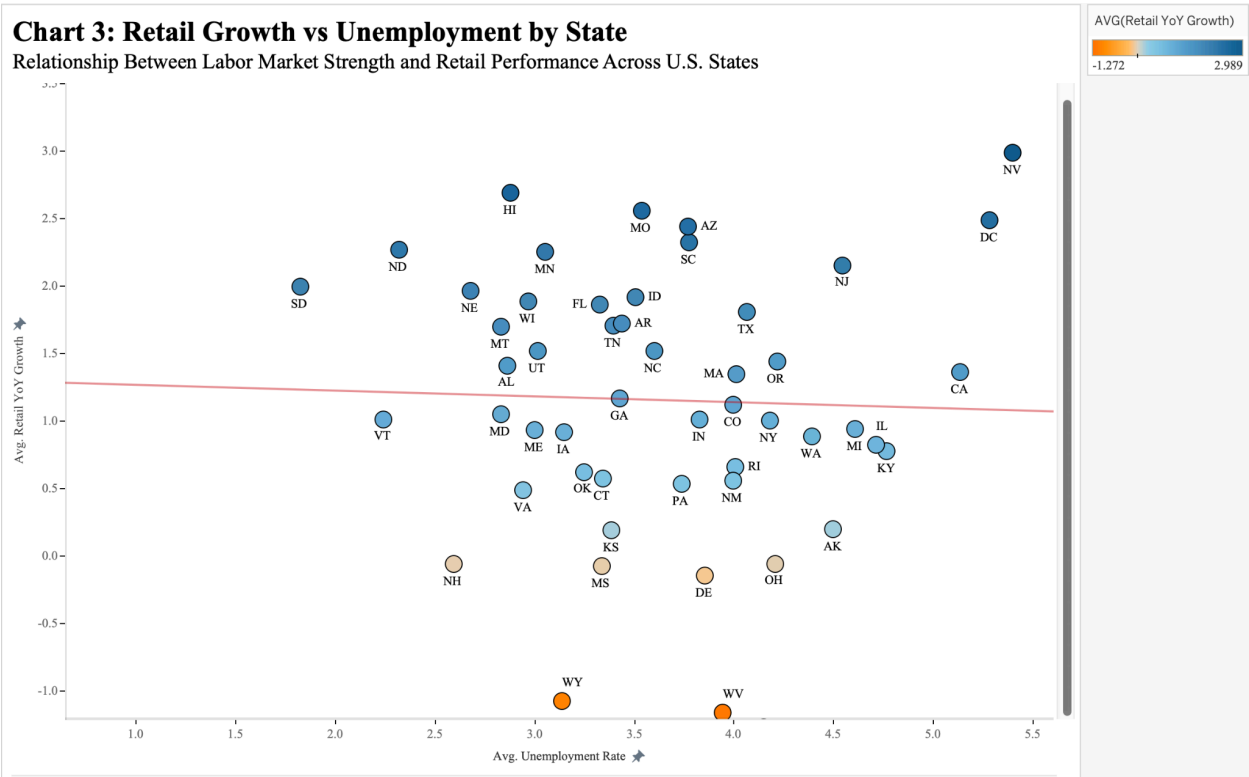
This chart compares average retail growth across major sectors nationwide (2023–2025) to identify which industries are driving overall retail performance. Retail Growth % is calculated from Census retail sales data and averaged across all states to provide a standardized national benchmark. By encoding growth using both bar length (position) and color (diverging gradient), the chart clearly separates high-performing sectors from those experiencing decline.

The results show strong growth in Health & Personal Care and Miscellaneous Retailers, while sectors like Gasoline Stations and Building Materials exhibit negative trends. This highlights a shift in consumer spending toward essential and service-oriented retail categories, while more volatile or price-sensitive sectors lag behind.

This chart complements the geographic analysis by explaining *why* certain states perform well—linking regional success to the strength of specific retail industries.

Limitations:
Because the data is aggregated at the national level, it masks regional variation in sector performance. Additionally, averaging across 2023–2025 may smooth over short-term economic shocks or recent changes in consumer behavior.

CHART 3:



Variables: Avg Retail Growth % (year-over-year), Avg Unemployment Rate, State

This scatterplot examines the **relationship between labor market strength and retail performance across states (2023–2025)**. Retail Growth % is derived from Census data, while Unemployment Rate comes from the BLS, allowing for a direct comparison between consumer spending and labor conditions. Each point represents a state, with position encoding unemployment (x-axis) and retail growth (y-axis), and color reinforcing growth magnitude.

The slight downward trend line suggests a **weak negative relationship**, indicating that states with lower unemployment tend to experience slightly higher retail growth, though the relationship is not strong. Several states outperform this trend, achieving high retail growth

despite moderate unemployment, while others underperform even with relatively strong labor markets. This highlights that unemployment alone does not fully explain retail success.

This chart deepens the analysis by connecting economic conditions to retail outcomes, helping identify states where strong consumer demand exists alongside favorable labor markets.

Limitations:
The relationship observed is correlational and does not imply causation. Additionally, averaging values across 2023–2025 may mask short-term economic fluctuations, and other factors (such as income levels or inflation) are not included in this analysis.

CHART 4:

Chart 4: Retail Opportunity by State and Sector
High-Growth, Low-Unemployment Markets Across U.S. Retail Sectors

AVG(Opportunity Score)
-9.69 10.41

Stateab..	Building Materials St..	Clothing Stores	Electronics Stores	Food & Beverage St..	Furniture Stores	Gasoline Stations	General Merchandis..	Health & Personal Car..	Miscellaneous Retailers	Motor Vehicle & Parts Deal..	Sportin Goods / H
SD	-1.56	3.33	-1.95	-1.61	-1.09	-3.94	0.15	8.01	10.41	0.37	-9
ND	-2.50	1.43	1.89	-1.69	-7.59	-6.40	2.83	5.33	9.03	4.14	-7
HI	3.53	3.06	-4.21	-7.23	-5.17	-1.56	-2.34	0.61	4.41	2.80	-0
NE	-5.21	-0.55	2.37	-3.18	0.56	-5.93	0.40	2.96	6.95	2.26	-6
MN	-4.04	-1.03	-1.04	-0.46	-2.84	-6.91	0.37	8.53	2.00	1.28	-5
MO	-5.68	0.80	-0.46	-2.27	-5.49	-6.21	-0.37	2.44	9.69	1.59	-4
WI	-3.10	-0.87	-5.74	-0.38	-5.21	-3.95	0.93	3.98	5.62	2.09	-6
MT	-4.44	-4.09	-5.21	1.16	-3.34	-7.97	-2.23	10.28	7.47	0.62	-5
VT	-4.49	0.76	-0.99	2.63	-5.99	-9.69	0.93	7.02	0.36	-5.54	1
AZ	-3.77	1.04	-0.42	-1.69	-1.45	-6.91	1.06	1.44	0.28	-1.19	-2
SC	-4.72	2.10	-7.97	1.00	-2.11	-7.91	-3.10	4.24	0.06	4.29	-2
AL	-2.57	3.69	-3.79	1.24	-5.22	-7.35	-1.65	1.03	4.49	-1.54	-3
FL	-3.30	2.10	-4.57	-0.94	-4.52	-9.27	-0.90	2.05	3.87	0.65	-2
UT	-2.77	-0.50	-4.92	-0.73	-0.99	-7.80	0.94	0.46	-5.61	1.30	0
ID	-2.90	1.24	-2.26	-2.37	-6.39	-7.13	2.51	1.49	0.42	0.99	-3

Variables: Opportunity Score (Retail Growth % – Unemployment Rate), State, Sector

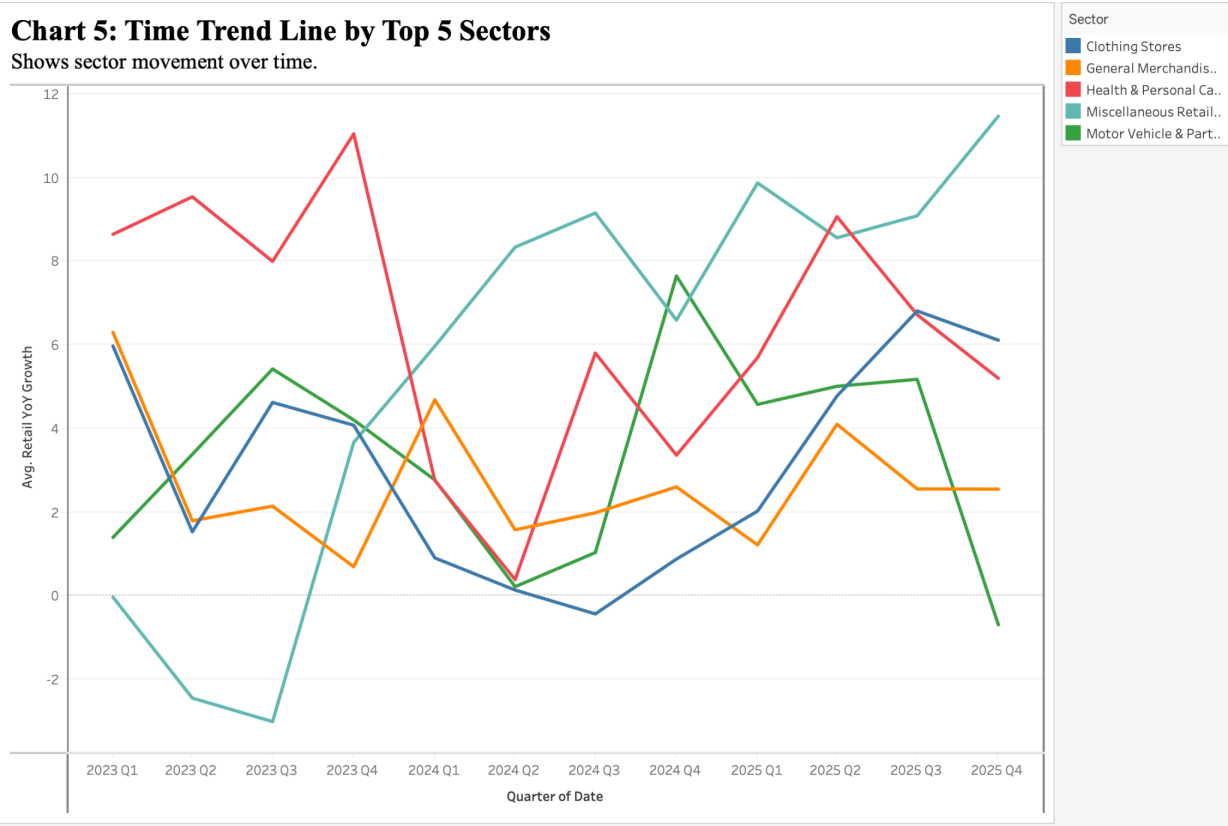
This heatmap identifies **high-opportunity retail markets by combining sector performance with labor market conditions** across states (2023–2025). The Opportunity Score is a derived metric that captures where strong retail growth coincides with lower unemployment, signaling favorable conditions for business expansion. Each cell represents a state–sector combination, with color encoding the magnitude of opportunity (blue = high, orange = low).

The chart reveals that Health & Personal Care and Miscellaneous Retailers consistently show high opportunity scores across many states, while sectors like Gasoline Stations and Building Materials tend to underperform. It also highlights that opportunity varies significantly not just by state, but by industry within each state, reinforcing the importance of targeted investment rather than broad geographic expansion.

This chart synthesizes the previous analyses by directly identifying *where and in which sectors* businesses are most likely to succeed, making it the most actionable component of the dashboard.

Limitations:
Because the Opportunity Score is a simple difference metric, it assumes equal weighting between retail growth and unemployment, which may oversimplify real-world dynamics. Additionally, the heatmap can become dense, making it harder to interpret smaller differences between sectors or states without interaction or filtering.

CHART 5:



Variables: Avg Retail Growth % (year-over-year), Quarter (time), Sector

This line chart tracks retail growth trends over time for the top 5 performing sectors (2023–2025), highlighting how sector performance evolves rather than relying on static

averages. Retail Growth % is derived from Census data and aggregated quarterly to smooth short-term volatility while preserving directional trends. Each line represents a sector, with position encoding growth over time.

The chart shows that Health & Personal Care and Miscellaneous Retailers maintain strong upward trends, indicating sustained consumer demand, while sectors like Motor Vehicle & Parts and Clothing exhibit more volatility. This suggests that some sectors offer more consistent growth opportunities, while others are more sensitive to economic cycles or seasonal demand.

This chart adds a temporal dimension to the analysis, helping determine whether high-performing sectors are consistently strong or experiencing short-term spikes, which is critical for long-term investment decisions.

Limitations:
Quarterly aggregation may still mask short-term fluctuations, and focusing only on the top 5 sectors excludes potentially emerging industries that may become important in the future.

Summary & Conclusions

Project 2 reveals that retail performance across the U.S. is not only geographically uneven, but also highly dependent on which sectors are driving growth and how those sectors interact with labor market conditions. We find that sectors like Health & Personal Care and Miscellaneous Retailers consistently outperform others, and that the strongest opportunities emerge where high retail growth aligns with relatively low unemployment. At the state level, several standouts emerge: **Nevada** shows the highest overall retail growth, indicating strong consumer demand, while **Louisiana** stands out as the strongest overall opportunity state when balancing growth with labor market conditions. States like **Texas and New Mexico** perform well in Clothing Stores, while much of the Midwest and Southeast—such as **Iowa, Minnesota, and Georgia**—are driven by Health & Personal Care growth. At the same time, weaker or more volatile performance appears in parts of the Northeast and Midwest, reinforcing that the U.S. cannot be treated as a single retail market.

Compared to Project 1, our work moves beyond identifying where retail growth is occurring to explaining why it is happening and where the best opportunities exist. By incorporating sector-level analysis, time trends, and a derived Opportunity Score, we provide a more actionable, multi-dimensional view of the retail landscape that Project 1 could not capture.

Using advanced Tableau features such as interactive maps, calculated fields, heatmaps, and integrated dashboards allowed us to connect geography, sector performance, and economic conditions in a single cohesive story. These tools made it easier to identify patterns, such as Nevada's high growth or Louisiana's strong opportunity position, and compare them across multiple dimensions. If we were to extend this further, we would add more interactivity, such as

dynamic filtering by time period or sector, and potentially incorporate predictive elements to forecast future growth.

Some data limitations from Project 1 remain. The analysis is still limited to state-level aggregation, which masks variation within states (e.g., differences between urban and rural markets), and it does not account for additional economic drivers such as income, inflation, or population growth. While we improved the dataset by adding sector-level detail and deriving new metrics like Opportunity Score, these broader influences are still not captured.

If the project were to continue, we would explore incorporating additional datasets such as median income, population trends, and e-commerce activity to build a more complete model of retail success. We would also consider moving to a more granular geographic level (such as city or county) and applying predictive modeling to identify emerging high-opportunity markets before they fully develop.